

An Adaptive Intrusion Detection System in Industrial Internet of Things(IIoT) using Deep Learning

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Abstract— The Industrial-Internet-of-Things (IIoT) is a product of the extensive use of the Internet-of-Things(IoT) in vital industries including manufacturing and industrial production. To improve industrial and manufacturing processes, the IIoT integrates sensors, actuators, and smart tools that can interact with each other. There are many advantages to IIoT for service providers and customers alike, but privacy and security are still major concerns. Cyberattacks in such a network have been reduced with the use of an intrusion detection system (IDS). Nevertheless, several current IIoT intrusion detection systems (IDS) suffer from issues such as an incomplete list of the network's attack kinds, an excessive number of features, models constructed using outdated datasets, and an absence of attention to the issue of imbalanced datasets. Our proposed intelligent recognition system can spot cyberattacks in IIoT-networks, which will help with the difficulties. Singular value decomposition (SVD) is employed by the suggested model to decrease data characteristics and enhance detection outcomes. If we want to keep our classifications from being biased due to over-fitting or under-fitting, we employ the SMOTE method. Data has been classified using a number of deep learning and machine learning techniques for both binary and multi-class purposes. We test the suggested intelligent model on the ToN_IoT dataset to see how well it performs. With the suggested method, we were able to achieve a 99.98% accuracy rate and a lowered error rate of 0.016% for multi-class classification, and a 0.001% reduction in the error rate for binary classification.

Keywords— IIoT, Intrusion detection, Cybersecurity, deep learning

I. INTRODUCTION

New business process technology is being propelled by the ever-evolving and exploited Internet of things (IoT) technology, which allows for communications and interactions among multiple devices via a network. As a result of the exponential growth of cybersecurity threats, numerous issues have emerged in various areas [1], including finances, credibility, enforcement, and corporate operations. Internet of Things (IoT) data storage is a common use case for cloud computing, which is based on a paradigm that provides customers with a range of resources and services as needed [2]. Typically, cloud computing reduces the amount

of human interaction between service providers and their customers [3]. Organizations and users alike have taken notice of its remarkable features. Moving from on-premises to cloud computing, however, could present several challenges, particularly in the areas of operation mechanism and security. The sensitive information kept on distant servers is the source of cloud computing's security risk. Because of this security risk, it attracts a lot of hackers and invaders, which discourages a lot of people from using or moving to the cloud [4]. The exponential growth of cyberattacks in recent times can be attributed to a number of factors. The availability and simplicity of hacking tools is a major factor; these enable inexperienced hackers to swiftly breach cloud storage without specialized training or expertise [5].

As a result of the dramatic increase in attacks targeting the Internet of Things (IoT), academics have recently focused on the privacy and security features of this technology. The Internet of Things (IoT) presents unique challenges when it comes to defending against these types of attacks compared to traditional networks. This is due to the fact that IoT devices are diverse, often underpowered, and have limited resources, which further complicates the matter [6]. The lack of regulation and clarity on device ownership and custodianship is another critical component of IoT networks that raises questions about who is liable in the event of an assault on the availability, confidentiality, or integrity of data or information [7].

Security and privacy concerns, such as confidentiality, integrity, and availability, are paramount at the first stages of designing a smart system's architecture, right up there with the information stowage and supervision challenges posed by the Internet of Things (IoT). Multiple security threats target IoT networks, putting users' personal information at risk and jeopardizing the networks' primary purpose of data storage and communication [8]. Denial of Service (DoS) attacks, for example, can take down an entire network or just a few nodes, preventing data from flowing freely. In order to breach various parts of the IoT network's security, attackers can

utilize scanning, enumeration, and reconnaissance attacks to gain sensitive user data and thereby threaten user privacy [9].

Significant financial and operational losses are caused by cyberattacks, which continue to pose serious security & privacy risks in smart settings. After a WannaCry virus version propagated over the manufacturing network, one of the biggest pure-play semiconductor businesses in the world reportedly shut down some of its fabrication facilities. A

substantial loss of almost \$170 million is the consequence of such an attack. Attackers gained unauthorized access to the control system in December 2015 [10]. Such intrusions into critical industrial control systems result in substantial downtime and monetary loss. A hacking scenario on IIoT networks is shown in Fig. 1.

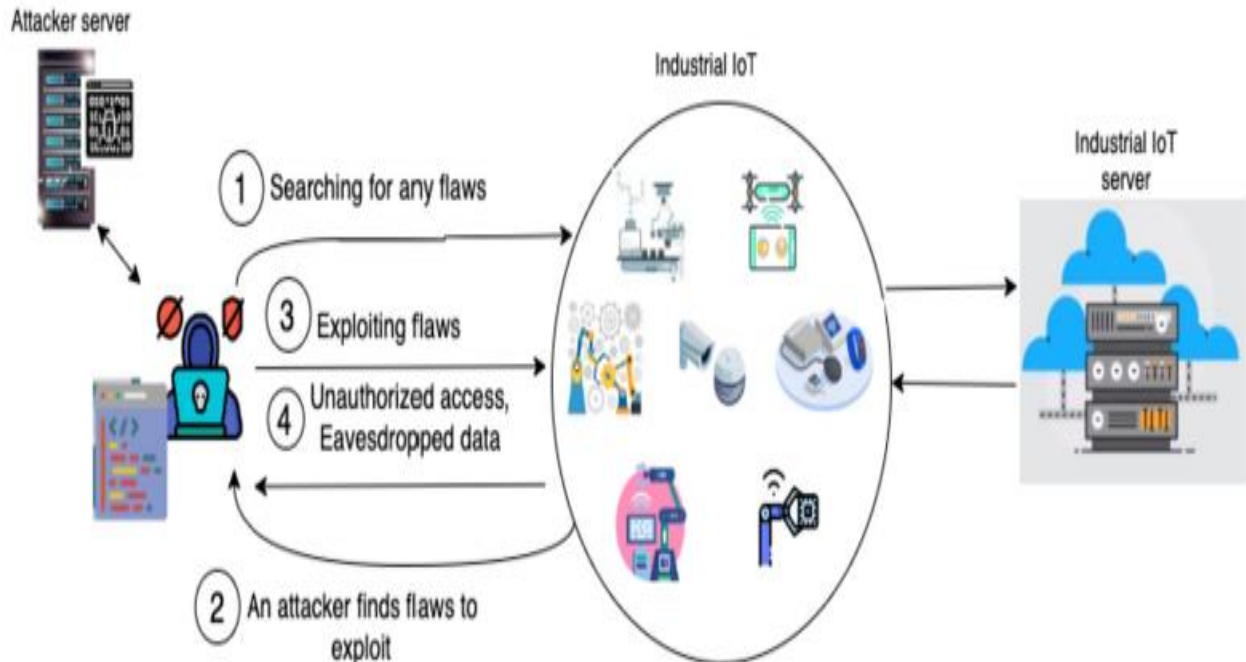


Fig. 1. Cyber-attacks structure in IIoT

Most crucially, in a real-world IIoT setting, the models won't perform as well as advertised. Furthermore, addressing imbalance through data pretreatment and data augmentation procedures is implicitly dependent on the model's performance. When it comes to data imbalances, the literature suggests that cost-sensitive learning methods in data mining are better at addressing them than basic data pretreatment and augmentation approaches. Instead of using data pretreatment and data augmentation to deal with the imbalanced WUSTL EHMS 2020 dataset, this work suggests a cost-sensitive learning technique. By combining an attention model with many deep learning layers, we may learn the network and patient biometric features that are most relevant for attack detection. Here are the key points of the proposed project:

- In order to successfully extract the robust spatial and time-series aspects of network flow and patient biometrics, the proposed model utilizes both CNNs and LSTMs. This approach is based on deep learning and is intended for use in an IIoT environment for network-based intrusion detection systems.
- Significant characteristics can be extracted from CNN & LSTM layers with the help of the suggested model's integrated global attention.

- An evaluation of the previously conducted research on intrusion detection in IIoT environments in relation to the suggested model.

Here is how the remainder of the paper is structured. Part 2 delves into works that are relevant. Section 3 lays out the approach and methods that are suggested. The experimental data are presented and discussed in Section 4, along with a comparison to related models. An overview of the previous work and some recommendations for further study make up the paper's last section.

II. RELATED WORKS

The use of ML & DL techniques to identify IoT network intrusions has been the subject of much academic study. An intrusion detection system (IDS) that uses deep learning to categorize network traffic is introduced. A strategy to detecting DoS attacks in IoT networks was published by Alenezi et al. (2021) using machine learning and deep learning [11]. If you want the best intrusion detection possible, the accuracy and precision are improved by the proposed effort. When it comes to identifying network intrusions in the IoT, that relies on classic machine learning methods rather of deep learning techniques. The author transformed PCAP files into log files using the Zeek tool and the CICIDS2017 dataset. To begin identifying network attacks, the author employs two sets of features. There are ten features in a simple feature set and fifty features in a complicated feature set. The author [12] opted for a more simplified set of features to increase speed after realizing that

the results from both sets were almost identical. Four popular datasets were transformed into NetFlow in the study. One of the most used protocols for collecting network traffic is NetFlow [13].

For both binary and multiclass classification, random forest was employed. On each of the four datasets, the binary classifiers outperformed the other two. On other datasets, though, the classifier failed miserably. The writers opted to boost performance by adding more features. In a related study, introduced an IDS for IoT devices that relies on deep learning [14]. They used a variety of network-based datasets to conduct a rigorous evaluation of DL-strategies for IoT settings. Another study from 2021 detailed a method for detecting Botnet assaults that relied on deep learning and the CNN network. They asserted that their approaches yield 99.04% accurate outcomes with a forecast time of 0.34ms.

Each of the five research articles highlights a different area of cybersecurity and technological growth by proposing a novel solution to a specific problem. In order to optimize hybrid microgrid systems, the authors of the article [15] provide TA-MaEA, an improved evolutionary algorithm that takes into account power supply, dependability, cost, and emissions. When contrasted with current algorithms, this method drastically cuts down on system expenses. An innovative HTA-GAN is proposed in study [16] for use in the IoT. This GAN is highly effective in detecting anomalies and is very robust, thanks to its BiGAN structure. To tackle the issue of security in networked control systems, introduces a protocol that is triggered by events and an algorithm that may be used for online predictive control. These systems are designed to keep running smoothly even when cyber-attacks or packet breakdowns occur. To enhance efficiency and dependability in decentralized IoT settings, the research [17] suggested an event-driven SOA-based situation-aware service coordination platform. Finally, a hybrid deep learning technique is presented in [18] to address Smart Grid cybersecurity. This approach achieves high accuracy in identifying DDoS assaults and enhances grid reliability. Taken as a whole, these studies show how far we've come in improving and protecting vital systems like Smart Grids, IoT infrastructures, and microgrids.

To tackle the problem of skewed data in the WUSTL EHMS 2020 dataset, the authors of the solution offered in [19] present a new method based on deep learning to manage intrusion detection in an IoMT context that is network-based.

Instead of using data pre-processing and augmentation, this strategy uses cost-sensitive learning, which is different from earlier efforts. During model training, this approach gives more weight to classes with fewer samples of IoMT network traffic data and less weight to classes with more samples. Network features, patient biometrics, and a mix of the two are some of the features used to evaluate the suggested model. It uses layers of LSTM & CNNs to reliably extract the time series and spatial aspects of patient biometrics and network flow. In order to help extract important features from the CNNs and LSTM layers, a global attention mechanism is also implemented. The study concludes with a comparison of the suggested model to previous intrusion detection tests conducted in an IoMT setting, demonstrating the approach's distinct advantages [20].

The paper [21] tackles the problems with recurrent neural networks (RNNs), such as their restricted scalability in LSTM and GRU variants caused by sequential computing, problems with fading gradients, and their inability to identify complicated properties of minority classes. To solve these issues and make the SRU design more accurate, the authors suggest adding features like bidirectional SRUs and skip connections. Next, we use the ToN-IoT dataset to evaluate this threat detection model's performance to that of GRU and LSTM RNN variations. The research goes on to suggest an understandable security model for IoMT network threat detection. The authors assert that their model is the first of its kind; it uses explainable AI (XAI) to let humans understand the logic and facts behind it [22].

III. PROPOSED METHODOLOGY

In Figure 2, we can see the structure of our intelligent-IDS. Parts of the suggested model include an IIoT dataset, a pre-processing step, feature-engineering, ML-based intrusion detection systems, and DL-based intrusion detection systems. The processing of data entails filtering, cleaning, and standardizing the data. The suggested approach uses SVD to do feature engineering by eliminating superfluous characteristics while preserving most of the crucial ones. In order to reduce the number of dimensions in a dataset, SVD can estimate the amount of variation that each individual vector explains [23]. In order to create the final models, various DL & ML-strategies are used during the train phase. The following subsections provide examples of the various parts of the suggested system.

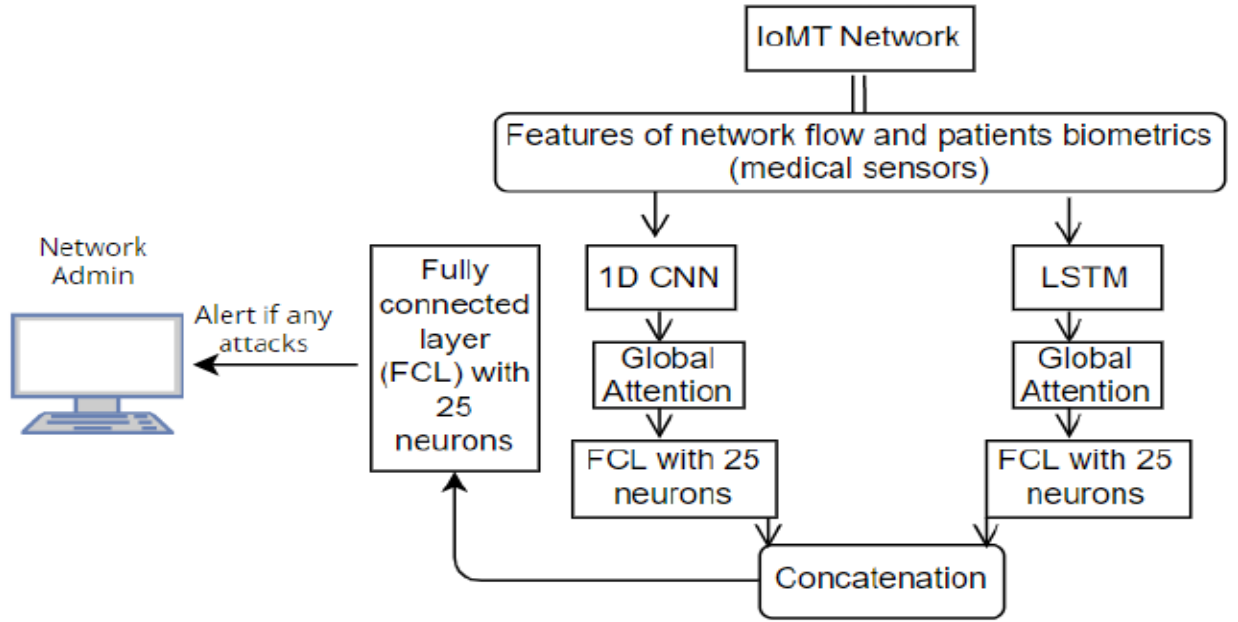


Fig. 2. Proposed model structure

To ensure that our ML and DL-based IDS uses the most relevant, consistent, and non-redundant characteristics, feature selection is an essential process. We keep most of the most important features while reducing the dimensions of the total matrix by removing superfluous ones [24]. Data storage, analysis, and processing are all made easier with smaller data sets that have fewer dimensions. The singular value decomposition (SVD) method is employed in our suggested detection model [25]. Since constructing models and scoring data usually takes more time when there are more columns. More columns usually mean more time spent creating models and scoring data, hence SVD helps minimize the dimensionality of datasets [26]. It could be time-consuming and reduce the model's performance quality when a column doesn't have any predictive value. Removing columns that seem to be collinear or that didn't demonstrate a lot of predictive value is a simple way to reduce dimension using an attribute priority ranking technique [27]. It is also possible to create new columns by using a linear combination of the existing ones. In every scenario, ML models can benefit from receiving the converted data set, which will enhance their performance

IV. RESULTS AND DISCUSSIONS

Here, we assess how well our designed detection approach performed. We evaluated the efficacy of our suggested intelligent-IDS using a number of performance indicators. We have tested and analyzed two different data classification levels: binary & multi-class.

Table 1. Performance Assessment of the proposed models for binary classification

Model	Acc(%)
Bi-LSTM	99.9
LSTM	99.7
GRU	99.8
BagTree	99.6
KNN	99.5

A. Binary classification: Table 1 shows that when comparing various approaches, DL-based IDS approaches like Bi-LSTM, LSTM, and GRU strategies perform better than the others. The accuracy rates of all the models are comparable, with the Bi-LSTM and GRU models achieving the best results in terms of error rate. Table 1 shows that DL-based IDS approaches take more time to train than ML-based IDS strategies [28]. It is well-known that GRU requires more time for training and testing than Bi-LSTM & LSTM strategies. We find that KNN classifier, which is regarded as a sluggish performer, has the longest testing time despite having the best training time. When compared to other models, GRU's lengthy training time is due to its heavy computation, but the DT classifier has the best testing time. For these reasons, we think that DL-based IDS models perform better than competing models with respect to Fig. 3's metrics for accuracy, precision, recall, F-measure, and AUC-ROC. While DL-based IDS models take longer to train and test, ML-based models get reasonable results.

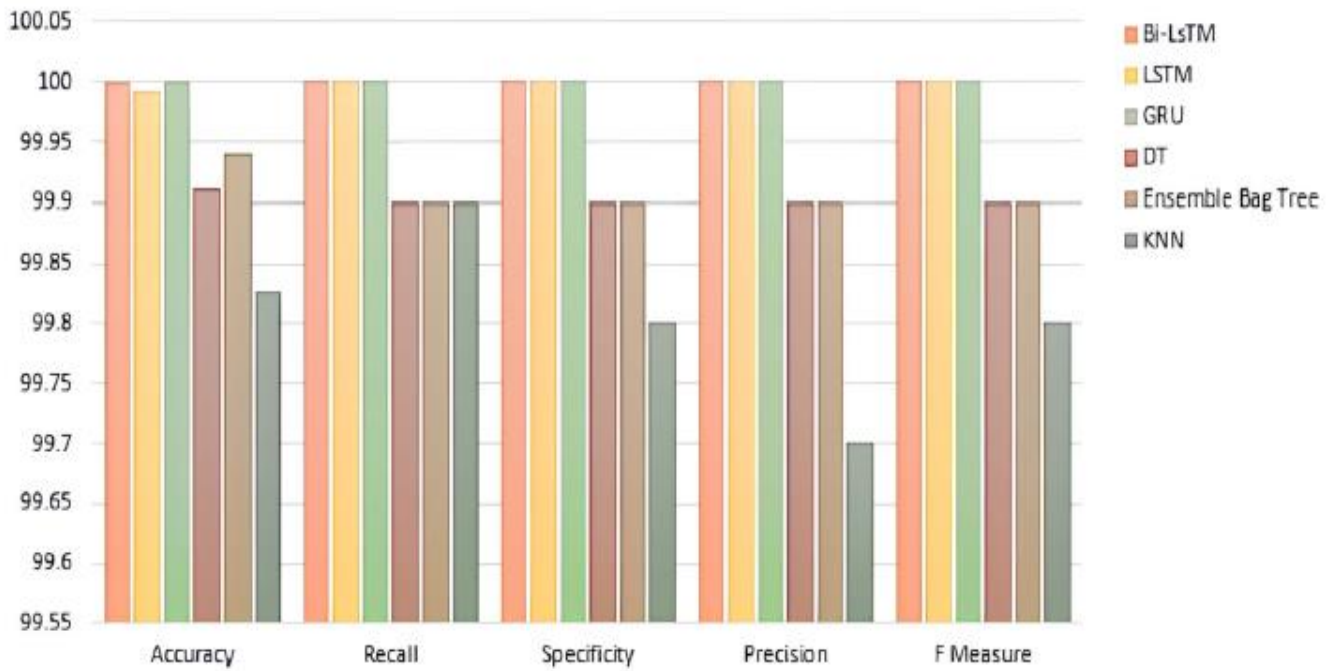


Fig.3. Performance Evaluation

B. Multi-Level class classification: Table 2 summarizes the investigational results of evaluating the proposed IDS models that are based on ML and DL. Table Figure 4 shows that while DL-based IDS approaches like LSTM, GRU, and Bi-LSTM all achieve comparable accuracy rates, GRU has the lowest error rate compared to the others. Although DL-based IDS models get somewhat higher accuracy rates, ML-based models like DT, ensemble bag tree, and KNN still manage to get respectable results [28]. The ensemble bag tree ML-based intrusion detection system model has the best error rate at 0.074. Training for DL-based IDS models takes a long time. The Bi-LSTM model takes the cake at 1207.2 seconds, easily the longest training period out of any of them. With the exception of Bi-LSTM, which takes longer than other DL-models owing to its intensive computation, all DL-based IDS models accomplish satisfactory testing times. In contrast, all of the ML-based IDS strategies train quickly, with the exception of the 143.457 second-taking ensemble bag tree. When comparing ML-based intrusion detection systems, the DT model has the fastest testing time at 0.065 seconds, while the KNN, which is a lazy classifier, has the longest testing time at 358.709 seconds. Though DL-based IDS models have better overall performance, they take more time to train and evaluate than ML-based IDS models.

Table2. Performance Evaluation for Multi-class classification

Model	Acc(%)
Bi-LSTM	99.9
LSTM	99.908
GRU	99.98
KNN	99.77

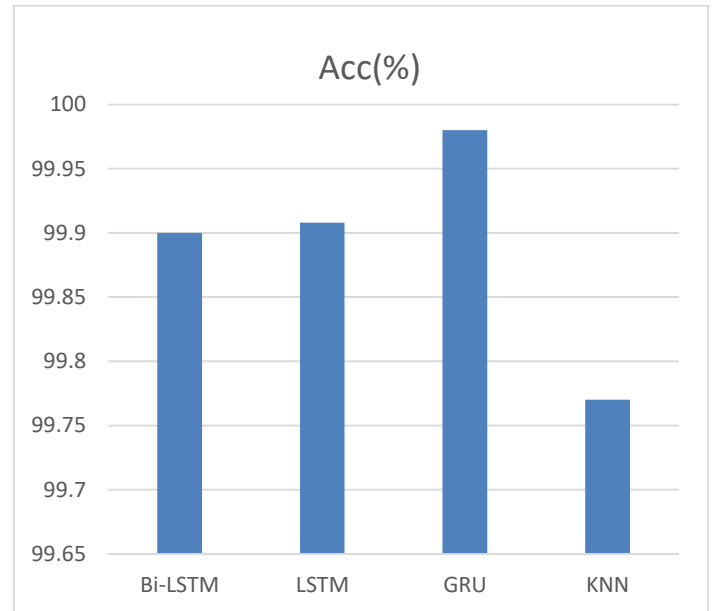


Fig.4. Performance estimation for multi-class

V. CONCLUSIONS

Protecting Industrial IoT from several forms of cyberattacks is the goal of this paper, which presents an intrusion detection strategy based on ML and DL. In order to improve the IDS's detection performance, the suggested mode applied the feature engineering approach known as singular value decomposition (SVD). To address the issue of biased classification caused by over-fitting and under-fitting in imbalanced datasets, we implemented the SMOTE method. In order to detect cyberattacks in such a networked environment, various intrusion detection systems (IDS) based on ML and DL are employed. For both binary and multi-class classification, a newly created IIoT dataset, ToN_IoT, was subjected to rigorous testing. We hope to train our algorithm on more diverse datasets in the future to enable it to detect a wider variety of cyberattacks. On top of that, we've integrated

a decision-making unit into our system so that it can respond appropriately to detected threats. In addition, we will look into other feature selection methods to reduce the detection models' training time.

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